

GetWild: A VR Editing System with AI-Generated 3D Object and Terrain

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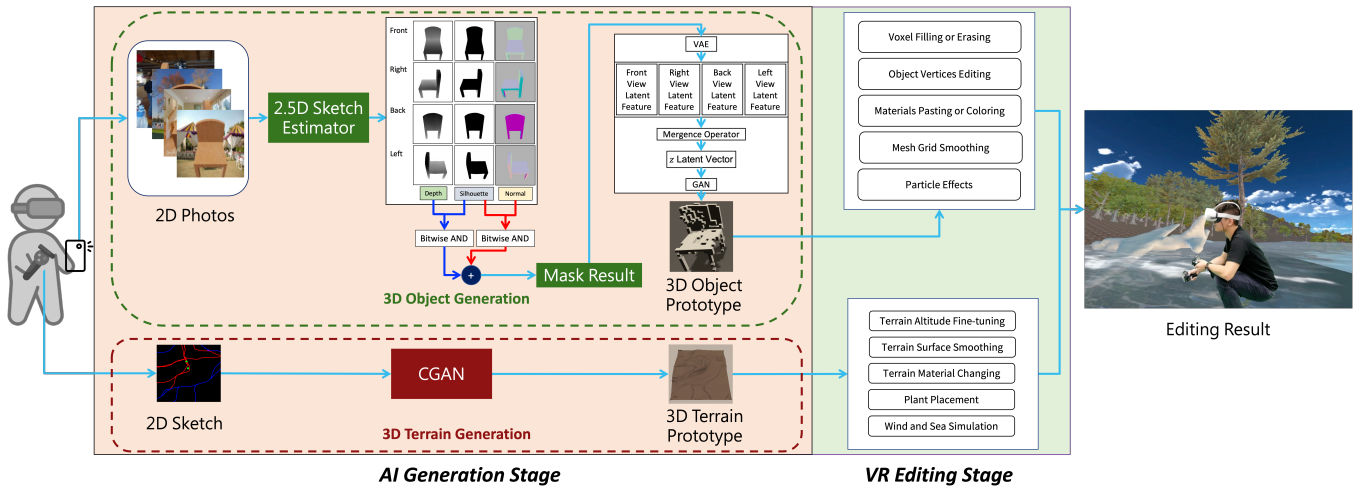


Figure 1: System framework of the proposed Getwild system.

ABSTRACT

3D environment artists typically use 2D screens and 3D modeling software to achieve their creation. However, creating 3D content using 2D tools is counterintuitive. Moreover, the process would be inefficient for junior artists in the absence of a reference. We develop a system called GetWild, which employs artificial intelligence (AI) models to generate the prototype of 3D objects/terrain and allows users to further edit the generated content in the virtual space. With the aid of AI, the user can capture an image to obtain a rough 3D object model, or start with drawing simple sketches representing the river, the mountain peak and the mountain ridge to create a 3D terrain prototype. Further, the virtual reality (VR) technique is used to provide an immersive design environment and intuitive interaction (such as painting, sculpturing, coloring, and

transformation) for users to edit the generated prototypes. Compared with the existing 3D modeling software and systems, the proposed VR editing system with AI-generated 3D objects/terrain provides a more efficient way for the user to create virtual artwork.

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1 INTRODUCTION

In the past, 3D environment artists typically use 2D screens and 3D modeling software to create various 3D content, which causes inefficiencies and challenges for the artists. As virtual reality (VR) technology has been proposed, some researchers indicate that performing 3D content creation in an immersive environment can improve the time cost of the modeling procedure. For example, Stark et al. [4, 8] suggested that creating 3D content using immersive VR systems involves less mental transformations than using traditional two-dimensional media. Mujber et al. [7] showed that VR is more intuitive for 3D content manipulation, speeding up the

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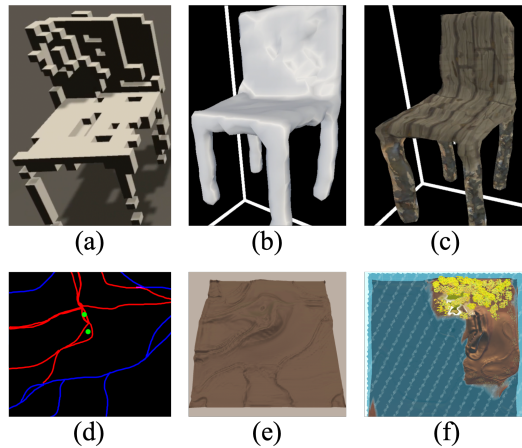


Figure 2: Example results of (a) the AI generated voxel representation of the 3D object; (b) the mesh representation of the 3D object; (c) and (f) the edited result created by using the proposed VR system; (d) the rough sketch representing the terrain features input by the user; (e) the AI generated 3D terrain.

design process. VR editing tools such as Tilt Brush [6], Blocks [5], and Ontlus [1] have been proposed recently. However, it is difficult for users without experienced 3D modeling skills to create artwork using these tools directly. In recent years, AI technologies have been widely employed in classification, regression, and generation tasks. Many works attempt to reconstruct 3D models from 2D references using AI generation models [11]. Inspired by these works, we develop GetWild, a system that combines an AI stage for 3D content generation and a VR stage for 3D content editing, as shown in Figure 1. The AI stage has two main modules: (1) **3D Object Generation** enables the user to create a rough object by capturing photos via a smartphone. The passthrough module [10] of the VR HMD is turned on so that the user can directly see what the smartphone captures ¹. (2) **3D Terrain Generation** allows the user to create rough 3D terrain with a few strokes depicting the main feature of terrain. GetWild provides rough 3D models for users to edit in VR environment, speeding up the 3D content creation pipeline.

2 SYSTEM FRAMEWORK

The GetWild system pipeline is illustrated in Figure 1. The proposed system is developed with Unity Engine to achieve intuitive editing interaction. The AI models are built and trained using PyTorch library. The AI stage allows the user to create rough 3D objects and terrain with simple input, and the VR stage provides a set of easy-to-use 3D editing interfaces so that the user can edit content details freely. More details of our system are described as follows.

2.1 3D Object Generation

In traditional 3D modeling workflow, 3D artists typically collect some photos of the target 3D object as references before modeling. The modeling time and quality rely on the artist's experience. For

users who are not well trained with enough modeling experience, we introduce a 3D object generation model based on the 2D images of the target object to facilitate the modeling process. The user can first capture four photos of the object from different views (i.e., front, right, back, left) via his/her smartphone. For each photo, 2.5D Sketch Estimator [9] is applied to predict the corresponding information of surface normal map (three-channel), depth map (one-channel), and silhouette map (one-channel). The surface normal map and depth map are masked based on the silhouette map and then concatenated together on the channel dimension as a single view masked result. Each single view masked result is then taken as the input of a 3D-VAE-GAN [12] to sample a latent feature in the correspondingly learned latent space. In order to obtain the best latent vector z representing a specific 3D object, we fuse the sampled latent features of the four multiview images (we average the four latent features in the current implementation). The fused latent vector z is then taken as the input of the GAN (decoder) to generate the voxel representation of the 3D object by classifying the coordinate as belonging to the object or not.

2.2 3D Terrain Generation

3D Terrain Generation enables the user to draw essential features of the terrain on the panel using the VR brush with different colors representing different semantics. Blue, green, and red represent the river, mountain peak, and mountain ridge, respectively. The drawing sketch can be considered as an RGB input image of the AI generator to obtain the prototype of 3D terrain. The AI generation model is constructed based on Conditional GAN (CGAN) proposed by Guérin et al. [2]. We use the dataset provided by [3] to re-train a CGAN which can model the mapping between sketch images and heightmaps.

3 RESULTS AND FUTURE WORK

Fig.2 (a) presents an example of a chair output generated by the AI 3D object generator. The generated voxel representation is not perfect, but the user can use the designed VR editing tools to fill or erase the object voxel, convert it to the mesh representation, edit the mesh vertices, paste material or color the object surface, smooth the mesh grid, and add the particle effects surrounding the object. Examples of the mesh representation and the editing result are shown in Fig.2 (b) and (c), respectively. Fig.2 (d) and (e) present an example of the sketch image and the corresponding terrain output generated by the AI model. The user can further edit the terrain by adjusting terrain height, smoothing terrain surface, changing the terrain material, placing the plant and setting the wind and water level. The editing result are shown in Fig.2 (f). More details about GetWild can be viewed at YouTube ². With the aid of AI and VR technology, users can not only obtain an AI-generated 3D object and terrain but also perform advanced editing in the VR environment to finely polish their creations. In the future, we intend to focus on improving the 3D object and terrain generation models, optimizing the system interface and user flow in accordance with user feedback, and building more advanced editing tools that can be used for collaborative creation.

¹Due to privacy issue, most HMDs do not allow the developer to capture images.

²<https://youtu.be/1NPLnT2YTeQ>

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